Data-Driven Prediction of Oscillatory AI-Training Behavior

Sign reversals, observable choice, sampling and probability calibration

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Abstract

This paper formulates oscillation forecasting as a target-specific probability problem. It defines a future increment-sign-reversal label, distinguishes that operational label from periodic dynamics, and develops causal WPM predictors with location, scale and optional structural context. Exact examples demonstrate loss/parameter differences, noise-induced reversals and sampling aliases. The mathematical treatment explains why probability accuracy can improve without a comparable ranking improvement and how class weighting and calibration affect interpretation.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 Causal observation and resolution contract

Let $\mathcal{F}_t$ contain the observations actually reported by issuance t: loss measurements, gradient and update statistics, known schedules and verified context. A delayed validation measurement belongs to the information set only when reported. The forecast horizon P is fixed by the query; W selects past extent, and m is a declared temporal sampling or internal-resolution parameter. These variables need not be functions of one another. For each target T, issue a forecast $p_{s,P,r}$

using only $\mathcal{F}_s$, with lens $r = (W, m)$. Once the label has matured, its loss can enter a causal surface

$$\widehat{L}_t(r; T, P) = \frac{\sum_{s \in \mathcal{A}_t} w_s \ell_T(Y^T_{s,P}, p_{s,P,r})}{\sum_{s \in \mathcal{A}_t} w_s}. \tag{1.1}$$

The same matured set $\mathcal{A}_t$ is used in numerator and denominator. Extend maturity when persistence or reporting delay requires follow-up beyond $s + P$. Unresolved negative outcomes are not known negatives. Empty history is missing risk, not zero risk. A pocket is a connected low-risk region under a specified adjacency and threshold. Retain each pocket's flow cross-sections, position, existence, signed errors and identity history. Fixed uses an eligible reference lens; Multi aggregates same-target alternatives; Auto changes a specified policy with causal context and matured utility. A pocket minimum is not an automatic event probability. An interpretable candidate event model is

$$p_{t,P} = \operatorname{logit}^{-1}\left(\beta_0 + \beta^\top z_t + \sum_i \alpha_i \phi(\mathcal{P}_i) + \sum_{i<j} \gamma_{ij} \psi(\mathcal{P}_i, \mathcal{P}_j)\right). \tag{1.2}$$

The coefficients, feature maps and any calibration map must be fitted from permitted data. Pocket labels must not have an artificial ordinal meaning; use a permutation-invariant aggregation or preserve a well-defined relational graph. Shared observations create dependence among pockets and cannot be counted as extra independent measurements. The compact archived AI representation used observable location/scale proxies in its multi-system scenario comparison, not explicit fits of all n/s parameters. A full scaling-spectrum branch is an optional extension that must calculate and qualify its own descriptors. Structural location/scale features and an actually fitted n/s spectrum are not interchangeable descriptions.

2 Specify the observable and the oscillation criterion

Oscillation can refer to parameters, gradients, training loss, validation loss or a fitted latent coordinate. These are not interchangeable. Let L_t be the specified reported loss, define future increments $\delta_j = L_{t+j} - L_{t+j-1}$ for $j = 1, \ldots, P$, and choose a zero/tolerance convention.

One operational statistic is

$$R_{t,P} = \frac{1}{P-1} \sum_{j=1}^{P-1} \mathbf{1}\{\delta_j \delta_{j+1} < 0\}, \quad P \geq 2, \qquad Y^{\text{osc}}_{t,P} = \mathbf{1}\{R_{t,P} > b_P\}. \tag{2.1}$$

The denominator is fixed here; zero increments count as no strict reversal. If increments below a tolerance are discarded, define the new denominator and handle empty eligible sets explicitly. A threshold learned from development labels is an operational boundary, not a universal physical constant.

The target is a property of an observed future segment. It is legitimate to use that segment to define a matured label, but no statistic calculated from it is an issuance feature. An alternative target might require spectral-band power or sustained autocorrelation; it needs its own labels, model and calibration.

3 Why sign reversals do not identify a physical oscillator

Proposition 3.1 (A periodic component can be hidden by trend)**.** *For sampled $L_j = \ell_0 + bj + A\cos(\omega j)$, increments satisfy*

$$\delta_j = b - 2A \sin(\omega/2) \sin[\omega(j - 1/2)]. \tag{3.1}$$

If $|b| > 2|A\sin(\omega/2)|$*, every increment has the same sign, although* $A \neq 0$ *gives a periodic component.*

Proof. Apply $\cos u - \cos v = -2\sin[(u+v)/2]\sin[(u-v)/2]$. The oscillatory increment is bounded in absolute value by $2|A\sin(\omega/2)|$, so it cannot reverse the sign of the larger trend b. □

Conversely, independent high-frequency measurement noise can generate many sign changes without coherent oscillatory dynamics. An operational reversal target is still useful if it matches the application, but its name must not imply more mechanism than the label establishes.

Worked example 3.2 (A sign-reversal worksheet)**.** For future losses $(1.0, 0.9, 1.0, 0.8, 0.9)$ including the current value, increments are $(-0.1, 0.1, -0.2, 0.1)$. All three adjacent pairs reverse, so $R = 1$. For $(1.0, 0.9, 0.8, 0.7, 0.6)$, $R = 0$. For $(1.0, 1.0, 0.9, 0.9, 0.8)$, strict reversal count is zero under the stated tie convention.

4 A noise reference and dependent reversal indicators

Let the signs of nonzero increments be independent with probability q of positive sign. A reversal probability is $2q(1-q)$. Thus the expected reversal fraction can be high even for an unstructured sign process.

Proposition 4.1 (Shared signs create dependence)**.** *For* $I_j = \mathbf{1}\{S_j \neq S_{j+1}\}$ *with independent signs* $\mathbb{P}(S_j = +1) = q$*, neighboring reversal indicators have covariance* $q(1-q)(1-2q)^2$*.*

Proof. Both reversals occur only in patterns $+-+$ or $-+-$, whose combined probability is $q^2(1-q) + (1-q)^2 q = q(1-q)$. Subtract $[2q(1-q)]^2$ and factor. □

At q=1/2, the open-chain flip indicators are independent fair binary variables: the initial sign and sequence of flips determine the entire sign sequence bijectively. Then $\mathrm{Var}(R) = 1/[4(P-1)]$. For other q, adjacent dependence changes this formula. In real loss measurements even the signs need not be independent; differencing shared noisy loss values itself creates temporal dependence.

This calculation is an analytical reference, not a universal null calibration. A reversal detector should account for trend, measurement precision and the task's relevant timescale rather than declaring every noisy fluctuation pathological.

5 Parameter oscillation and loss oscillation differ

For a scalar quadratic loss $L(\theta) = \lambda\theta^2/2$, fixed-step gradient descent gives $\theta_{j+1} = a\theta_j$, $a = 1 - \eta\lambda$. With $-1 < a < 0$, parameters alternate sign and contract, but $L_{j+1} = a^2 L_j$ decreases monotonically.

Worked example 5.1 (A damping oscillator invisible to a loss-reversal label)**.** At $a = -1/2$, the parameter sequence from $\theta_0 = 1$ is $(1, -1/2, 1/4, -1/8, \ldots)$. Loss scales by 1/4 at every step and its increments have one sign. A loss-increment reversal model is not wrong if it predicts no reversal; it was never asked to predict parameter sign alternation.

For momentum and multiple curvature modes, different observables may carry different frequencies and damping. The data-driven predictor does not receive the exact future optimizer law; these examples identify what the chosen telemetry can observe.

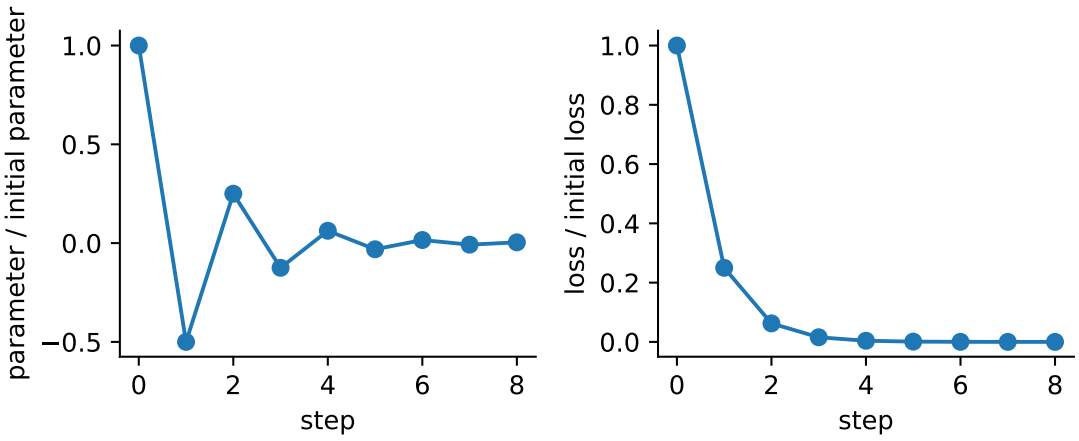


Figure 1: Exact quadratic contraction with multiplier -1/2 alternates parameter sign while loss decreases monotonically. The observable determines what an oscillation label can detect.

6 Sampling, aliasing and the meaning of m

Uniform samples of $\cos(\omega t)$ cannot distinguish all frequencies: at integer times, $\cos[(2\pi - \omega)j] = \cos(\omega j)$. A period shorter than the sampling interval can be completely aliased. Increasing interpolation density after observation does not supply missing information.

If m is sampling stride, coarser m changes the apparent increment signs and number of cycles retained inside W. A target defined on the original reporting clock must remain fixed while candidate lenses vary. Otherwise choosing m can change the label rather than improve prediction of a common event. If a low-pass or anti-alias filter is part of the measurement map, record its causal implementation and group delay.

7 Causal predictors and pocket histories

A location branch can summarize level, slope and curvature; a scale branch can summarize increment dispersion and its growth. Context can include known schedule phase, recent forecast revisions and past sign statistics. Pairwise pocket features can represent agreement or disagreement across resolutions, while keeping their flow curves, positions and existence intact.

Fit $p_{t,P} = \mathbb{P}(Y^{\mathrm{osc}}_{t,P} = 1 \mid \mathcal{F}_t)$ with a specified statistical model. No formula maps pocket depth alone to this probability. An Auto policy should not be activated solely because several local minima exist. Its update must use matured utility and preserve an eligible reference when evidence is sparse.

An upcoming known cyclic learning-rate schedule can be an issuance input. A future optimizer change chosen later is not known. Repeated sign flips under a deliberate schedule may be normal operation; the action policy needs a separate criterion for harmfulness.

8 Class weighting is not probability calibration

Binary classifiers are sometimes fitted with positive and negative class weights w_1, w_0. These change the optimum probability scale even with unlimited data.

Theorem 8.1 (Weighted log-loss optimum)**.** *If the true conditional event probability is p, minimizing weighted conditional log loss gives*

$$q = \frac{w_1 p}{w_1 p + w_0(1-p)}, \qquad \operatorname{logit} q = \operatorname{logit} p + \log(w_1/w_0). \tag{8.1}$$

Proof. Differentiate $-w_1 p \log q - w_0(1-p)\log(1-q)$ with respect to q and solve the zero-gradient equation. The second derivative is positive in the interior when both classes have positive weight. □

A fitted class-weighted score is therefore not automatically a calibrated event probability. Under this idealized model one can correct the log-odds offset. In practice misspecification and population shift require separate calibration data. Calibration decisions cannot use the unresolved current horizon.

9 Probability score, ranking and an explicit example

The binary squared-loss decomposition gives expected Brier $p(1-p)+(p-q)^2$. Strictly increasing recalibration preserves score order, and thus pairwise ranking, while changing Brier. At true p=0.2, scores 0.4 and 0.25 have expected Brier 0.20 and 0.1625, while both can preserve the same ranking against other cases.

For sequence-level oscillation, score the stated segment event. A pointwise probability of one reversal is not the probability that the reversal fraction exceeds b. Under a fully specified independent-flip model, that fraction has a binomial tail; under realistic dependence it generally does not. A direct event classifier or a joint path model can estimate the segment event without an independence shortcut.

10 Forecasting oscillation is not selecting a remedy

An oscillation forecast describes behavior under a specified continuation policy. It does not by itself establish that reducing learning rate improves future validation loss. Counterfactual remedies need an action-response model or an appropriate causal design. Healthy cyclic schedules and harmful divergence may have similar reversal fractions.

If the intended decision is harmful-oscillation intervention, define a joint target such as oscillation plus sustained loss deterioration, or use separate predictions with a cost model. Do not relabel the original sign-reversal event after seeing its outcomes.

11 Application worksheet and telemetry comparison

Specify the reporting clock, observable, increment tolerance, P and reversal-fraction threshold. Construct historical labels only after the complete future loss block is available. Fit same-target telemetry, structural and combined models using identical cutoffs. Build target-specific W,m landscapes from their actual issued losses, retain pockets, and calibrate the selected output with separate permitted labels.

Return probability, applicability, time units and the precise oscillation definition. Report an abstention if the observation clock cannot resolve the intended behavior. Refreshing after a new measurement creates a new issuance rather than a prediction made earlier.

Scoped evidence statement. In the archived compact AI comparison, the structural location-plus-scale branch improved oscillation Brier relative to the specified telemetry comparator, while ranking differences were not uniformly established. At the longest compared horizon, AUROC was close (approximately 0.655 versus 0.654) [4]. This supports the requested closeness statement for that metric and target, not universal equivalence or universal superiority. The branch used measured structural proxies and learned coefficients; it was not given the optimizer's true governing equation.

12 Rules and solved questions

1. Name the observable and criterion; periodic components, parameter alternation and loss reversals are distinct.
2. Fix the target clock before varying m. Do not let resampling change the event being predicted.
3. Preserve signs and dependence; ties and near-zero increments need a declared convention.
4. Separate probability calibration from ranking and class weighting.
5. Distinguish a forecast of behavior from a causal claim about a remedy.

Exercise 12.1. Can a strongly declining loss contain a periodic component and have zero sign-reversal fraction?

Solution. Yes. The trend-dominance proposition gives a sufficient condition: $|b| > 2|A\sin(\omega/2)|$.

Exercise 12.2. If q=1/2 and P=5 under the independent-sign reference, what are the mean and variance of the reversal fraction?

Solution. Its mean is 1/2 and variance $1/[4(5-1)] = 1/16$. This reference does not apply automatically to differenced correlated losses.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. This study shares the AI prediction collection DOI 10.5281/zenodo.23784369.

References

[1] T. Nguyen. *The WPM Framework: A Generalized Mathematical Architecture for Cross-Scale, Spatial, and Dynamical Prediction.* Zenodo, 2026. doi:10.5281/zenodo.23248983.

[2] T. Nguyen. *An n/s Scaling-Spectrum Framework for Generalized Moment Scaling: Curvature, Signed-Moment Asymmetry, and Prospective Moment-Order Validation.* Zenodo, 2026. doi:10.5281/zenodo.23111783.

[3] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 102(477), 359–378, 2007. doi:10.1198/016214506000001437.

[4] T. Nguyen. *Target-Conditioned Structural Forecasting of AI-Training Transitions: Frozen Multi-System Evidence for 1A–1D.* Version 1.1, October 2026. Archived manuscript, pp. 1–7. Evidence cited only for the scoped comparison statement.